

NexysA7 UART

By: Ben Robles

Cal Poly Pomona
Summer 2026

Design:

Idle	Start	D0	D1	D2	D3	D4	D5	D6	D7	Stop	Idle
1	0	1	0	0	0	0	0	1	0	1	1

Variable baud rate 8-N-1 UART protocol at 100 MHz system clock

Modules:

- baud_tick.v
- tx.v
- rx_sync.v
- rx.v

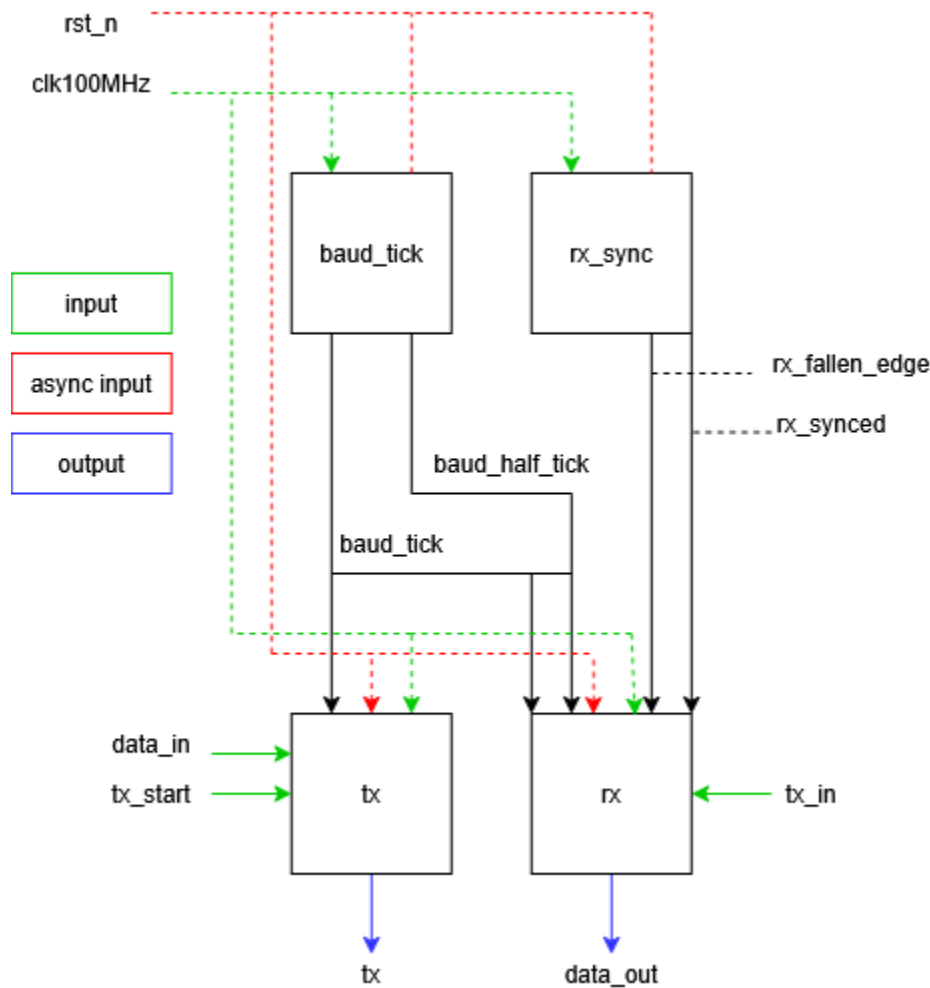
Explanation:

The design is implemented with an asynchronous reset

- Primary inputs: 100 MHz clock, rst_n, data_in, tx_start, tx_in
 - Primary outputs: tx_out, data_out
1. baud_tick.v generates a tick for each full and a half bit period
 2. tx.v uses a Moore FSM to drive tx each frame (10 bit periods) through Idle → Start → Data[0] → ... → Data[7] → Stop → Idle

3. rx.sync.v syncs the asynchronous tx_input to the system clock and detects falling edges (start bit)
4. rx.v uses a Moore FSM to read the tx_input frame in a similar way to tx.v. Each bit is sampled at the halfway tick of its bit period

Block Diagram:



Resources:

To build off of my digital circuit design verilog lab, I asked ChatGPT to generate a lab style write-up based on my professors previous labs. These were not followed strictly and mainly used as a starting point. AI was used for troubleshooting and learning but all code was written and verified by hand.

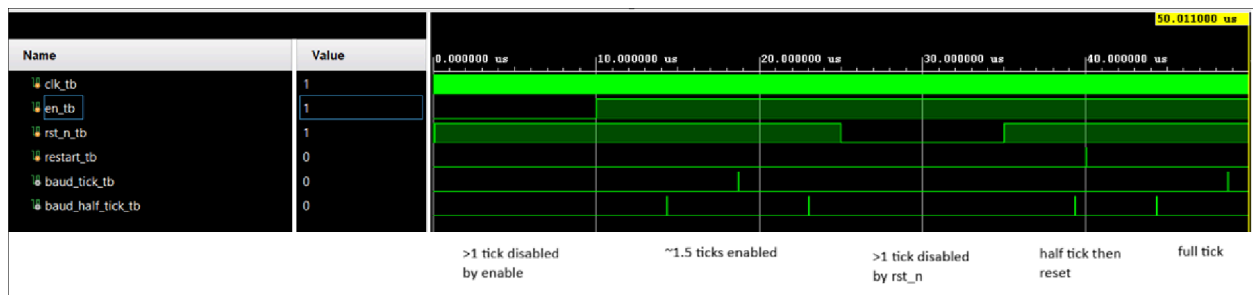
Simulation:

Testbenches for each module. rx_tb uses a loopback simulation.

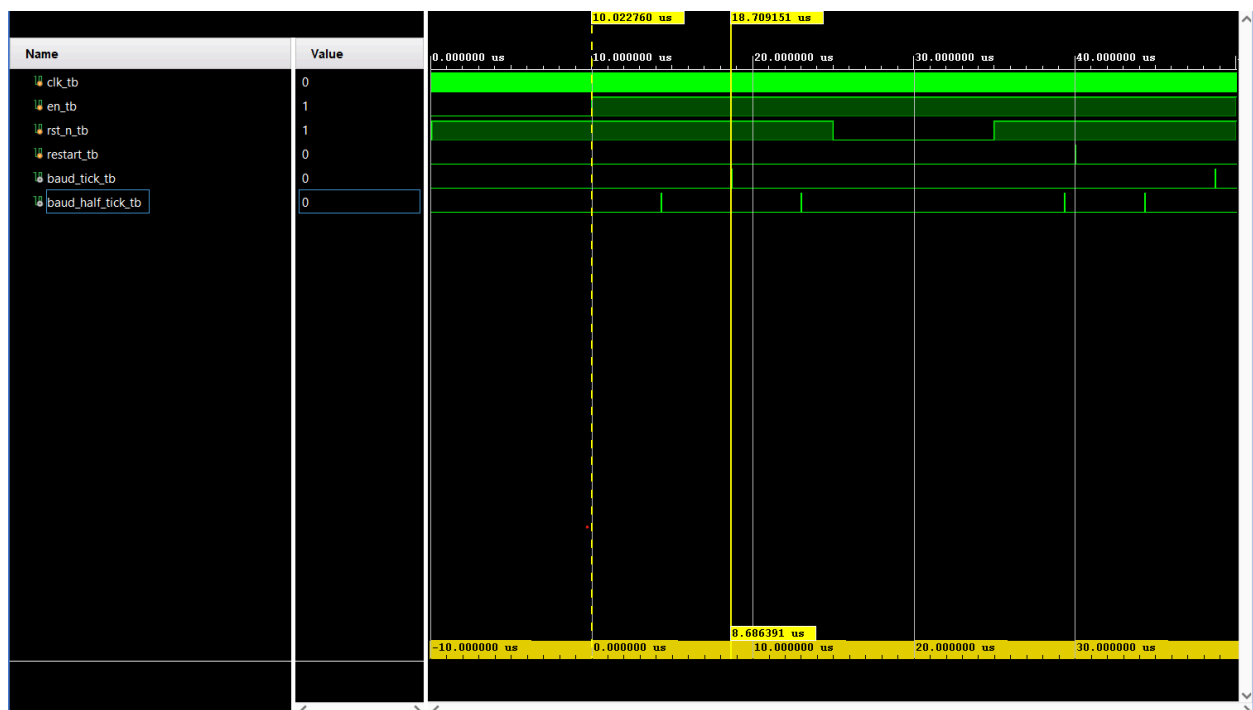
See waveforms folder for higher resolution annotated images

baud_tick.v:

full tick = system clock / baud rate = 100 MHz / 115200 = 8.68 μ s



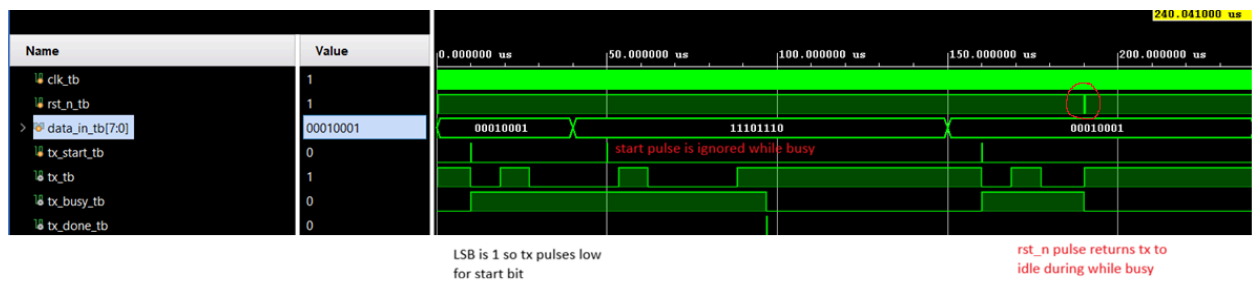
First full tick:



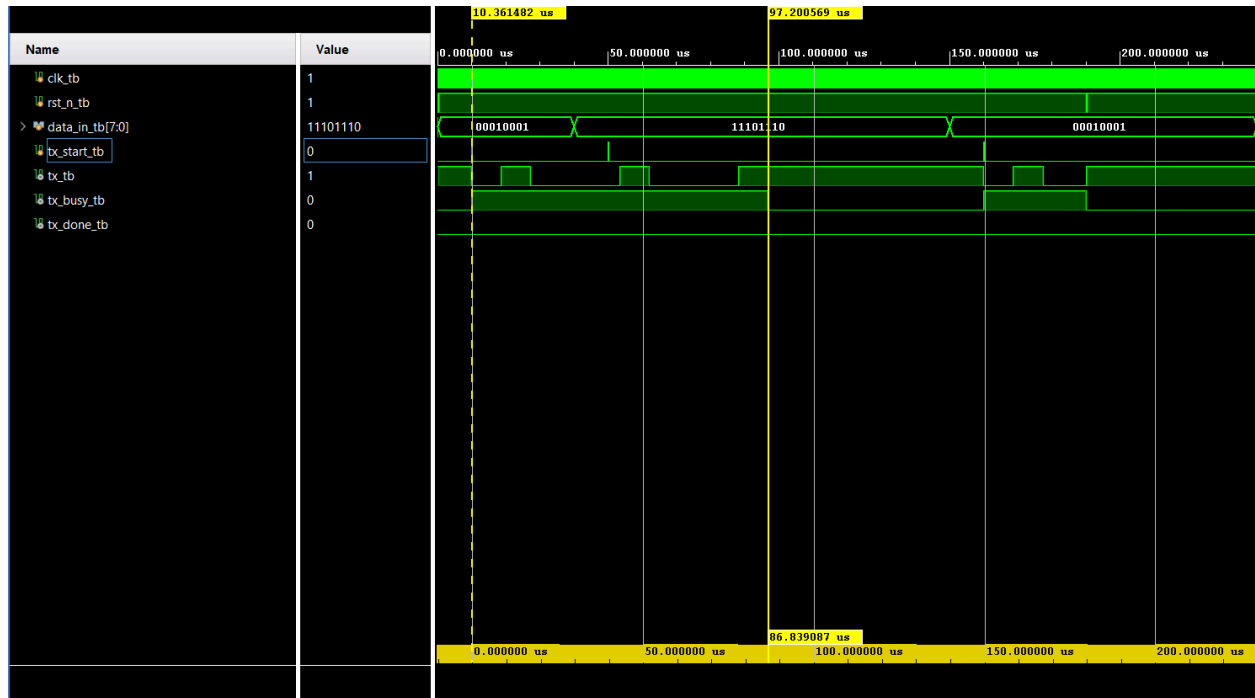
Full tick after final reset:



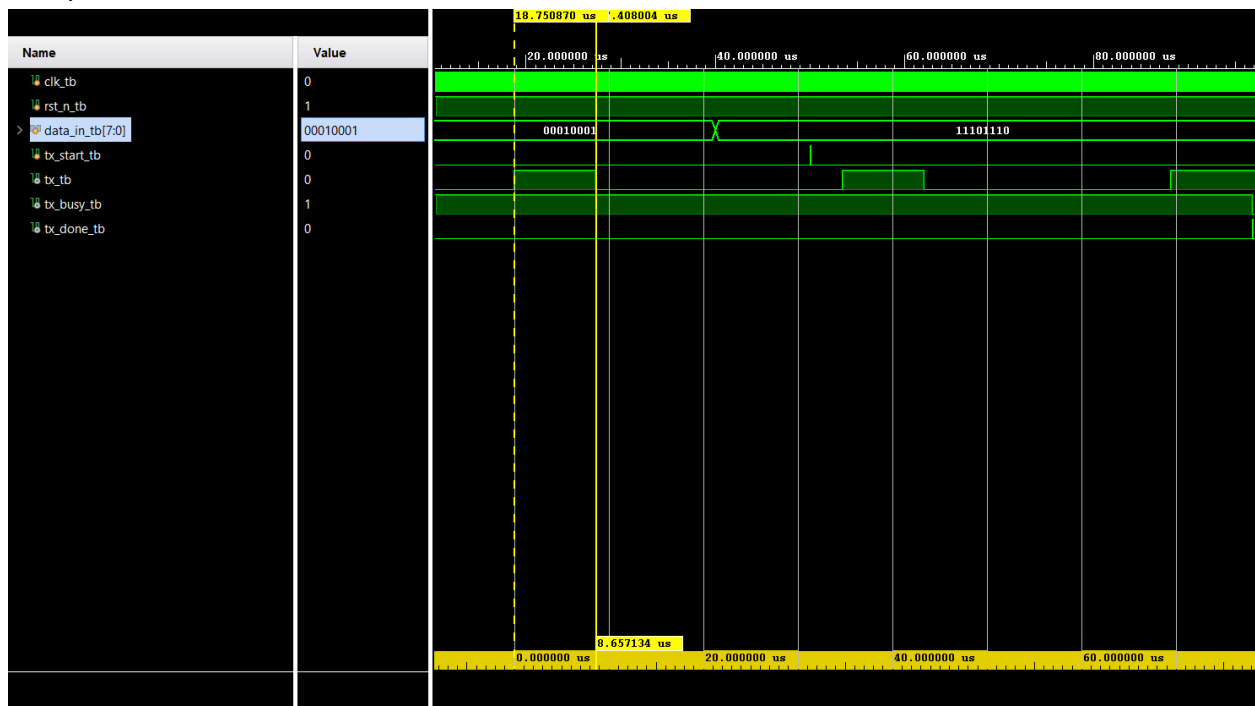
tx_tb.v:



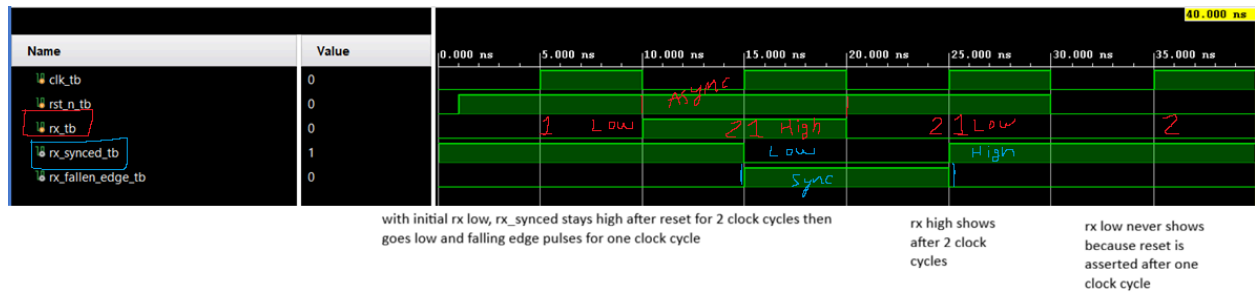
A full frame is 10-bits or 86.8 μ s:



1 bit period:

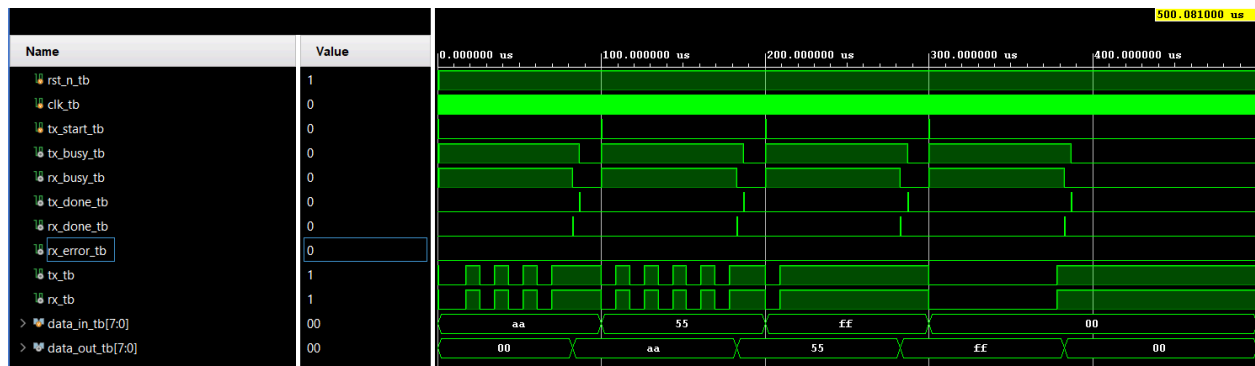


rx_sync_tb.v:



uart_tb.v:

Loopback test: data_in = data_out



One frame:



Conclusion:

I am away from the lab for the summer so I don't have the resources to perform a demo but it is planned for the future. This repo/project is intended as a framework for future projects.

I've used UART for years but never knew exactly how it worked. I enjoyed working on this as a first project into my pursuit of Verilog

(Revision 1: Ben Robles 8/12)